

WANG RUIJUE

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SUMMARY

Highly motivated, results-driven Data Science professional experienced in strategic AI implementation and workflow optimization. Proficient in requirements gathering, developing scalable ETL data pipelines, data warehouse architecture, and building data analysis models. Currently pursuing an M.Sc. in Digital Financial Technology at the National University of Singapore (NUS) and holding a B.Sc. in Data Science (Distinction). Proficient in Python and SQL. Experienced in ensuring data quality, consistency, and integrity across high-volume datasets (e.g., BigQuery). Skilled at applying quantitative reasoning to design systematic and reliable data solutions. Actively seeking an early-career role as an AI or Big Data Analyst/Scientist, aiming to leverage technology and Generative AI to drive digital transformation.

EXPERIENCE

Data Analyst (Engineer) Intern 05/2025 – 12/2025
NTUC Health

- Led client analysis projects by preparing and processing BigQuery data, applying advanced statistical methods (factor analysis, clustering), and delivering actionable insights via interactive dashboards to support business strategy.
- Developed 4 scalable DataMarts and multiple Looker dashboards with AI-powered features, streamlining reporting workflows and cutting ad-hoc requests by 70%, while empowering internal teams through targeted training sessions.
- Unified fragmented systems into a GCP-based ODS CDM ADS pipeline, improving data consistency by 40% and accelerating data retrieval by 60%, enabling faster business insights.
- Collaborated cross-functionally in Agile (Scrum) teams, using JIRA and Confluence to manage tasks and ensure alignment between business and technical stakeholders.

Data Analyst Intern 08/2024 – 04/2025
MEXC (Fintech)

- Served as the primary liaison for cross-functional teams, independently managing the referral platform's website updates, promotional design, and assisting with bonus reconciliation.
- Built Python + LLM-based automation pipeline for HR data audits, reducing manual effort by 12 hours/week with 99% accuracy
- Designed and deployed VBA-automated reporting robots on excel, cutting report preparation time by 60%

Data Management Intern 05/2023 – 05/2024
Asian Institute of Digital Finance, Credit Research Initiative

- Automated anomaly detection on 80,000+ firms' financials (MATLAB, Python), maintaining 99.5% accuracy and cutting FS validation time by 50% on 500k+ datasets using matrix-based approaches
- Integrated Bloomberg and ThomsonReuters data into MSSQL, achieving 100% coverage for downstream credit risk analytics and reporting.

Quantitative Strategy Research Intern 05/2022 – 08/2022
Beavo Investment

- Streamlined ML pipeline by preprocessing 2,000+ ZIP archives into standardized datasets, reducing modeling time by 70%, and halving backtest runtime via parallelization across Windows/Linux
- Engineered and optimized stock-return models using Qlib, feature selection, and regularization—achieving 15% lower MSE, 20% higher R², and a 12% Sharpe ratio improvement over legacy models

ACADEMIC PROJECT

Cloud-Native Financial Management Platform 09/2025 – 11/2025
FinanSync [Demo]

- Engineered a robust, multi-tenant data ingestion pipeline that uses a three-tier hybrid system (Historical Matching, Rule-Based Logic, Gemini AI Classification) to automatically categorize transactions from CSV uploads, achieving 80-90% automation.
- Developed a comprehensive analytics engine to deliver actionable insights including: Recurring Pattern Identification, Outlier Detection and a holistic Financial Health Score.
- Created a React-based SPA featuring secure AWS Cognito authentication and interactive dashboards, complemented by a Gemini-powered chatbot for SG-context natural language financial queries and summaries

FinTech Consulting Project 10/2025 — 11/2025
FinTech Industry and Venture Analysis

- Conducted in-depth analysis of established FinTech leaders (Stripe, Block) and digital banks

CERTIFICATION

NUS Science & Technology Undergraduate Scholarship (MOE Full Scholarship)
FRM Part I (Financial Risk Manager)
BDA Junior Certificate
Tan Ean Kiam Arts Awards 2022

LANGUAGES

English Advanced ●●●●●
Mandarin Native ●●●●●

TECHNICAL SKILLS

Programming: Python, SQL, Excel, VBA, MATLAB, R, PySpark (Self-Study/Basic)
Cloud/DevOps: GCP, AWS, Airflow, Git, Bitbucket, Docker
Databases: Metabase, Google BigQuery, SQL Server, MongoDB
ML/AI Models: LLMs/Generative AI, Deep Learning (LSTM, NLP), Logistic Regression, Random Forest, XGBoost, K-means, Factor Analysis, Statistical Modeling
Finance: Bloomberg Terminal, Risk Management, Portfolio Optimization, Financial Modeling
Visualization: Looker, Tableau, Power BI
Project Management: Agile, Scrum, JIRA, Confluence

CORE COMPETENCIES

- Agile Project Management
- Data Analysis & Business Strategy Support
- Generative AI & LLM Application
- Scalable ETL Pipeline Development
- Financial/ Credit Risk Analytics
- Data Visualization Dashboard & Reporting

INTERESTS

- NUS Chinese Drama Secretary
- NUS Stats & Data Science Society Pub
- Broadcasting & Event Hosting
- Chinese Opera Performance
- NUS Latin & Ballroom Dancing
- Social Media Marketing
- Sports & Traveling

- (MariBank) to evaluate business models, competitive positioning, and growth potential.
- Formulated a strategic product redesign for an existing consumer FinTech app to drive user and revenue growth, addressing operational considerations and market needs.
- Designed a new FinTech venture concept (KinVest) to address an unmet market need, developing a comprehensive pitch deck that included market sizing, competitive analysis, business model, and go-to-market strategy.

Blockchain Project

01/2025 – 04/2025

IntelliDeFi Invest

- Built real-time crypto price pipelines and LSTM forecasting models, reducing data latency by 40% and enabling smart contract-driven investment decisions
- Developed risk-based asset allocation logic integrating ML signals with investor profiles, improving simulated Sharpe ratio by 15%

Machine Learning Project

01/2025 – 04/2025

Real Estate Revenue Prediction

- Aggregated and cleaned NYC real estate and financial data to build ML-ready datasets for revenue and EBITDA prediction; engineered domain-specific features and conducted EDA to identify drivers of profitability and detect data anomalies
- Developed and fine-tuned LSTM models within a stacked ensemble (with tree-based models), achieving a 25% performance uplift over single-model baselines in terms of RMSE and R^2 .

Text Mining Project

03/2025 – 04/2025

No Regret to Social Media Post [Demo]

- Developed the core Offensive Language detection pipeline, leveraging SVC, LightGBM, and LSTM models on TF-IDF features, including SMOTE data balancing and hyperparameter tuning.
- Designed a modular, four-part risk detection framework combining regex (for PII/financial data), DistilBERT (for nuanced hate speech), and BERT Zero-Shot Classification.
- Achieved sub-50ms inference latency on a GPU-accelerated pipeline, ensuring real-time moderation capability for content fraud/risk proxy detection.

Full-Stack AI/Chatbot Project

10/2022 – 11/2022

AI Assistant App for Medical Sales Rep

- Conducted requirements analysis for a full-stack conversational AI chatbot designed to assist medical sales representatives with complex product queries, focusing on improving their access to information.
- Engineered the chatbot core with answer retrieval, confidence scoring, and alternative suggestions to ensure high utility in healthcare workflows.
- Executed end-to-end usability testing in a simulated clinical setting, validating its impact on efficiency.

Financial Analysis & Data-Driven Valuation

08/2024 – 04/2025

Case Analysis and Robo Advisor

- Capital Project Modeling: Evaluated two mutually exclusive hotel projects using NPV, IRR, and EAA; recommended the in-house pub option with a projected NPV of 1.60M baht and an IRR of 36.95%.
- Relative Valuation: Conducted a relative valuation of General Motors (GM) against peers (F, RACE, STLA) using a Python-based data collection pipeline (yfinance API).
- Investment Recommendation: Demonstrated GM's undervaluation using the industry average EV/EBITDA ratio (16.67), reverse-engineering a target Market Cap of \$315.79 billion (vs. actual \$50.37 billion), and advised a strategic long position/
- Built Excel-based robo-advisor integrating fund diversification, risk profiling, and Efficient Frontier optimization to deliver personalized portfolio strategies

EDUCATION

Master of Science in Digital Financial Technology

08/2024 – 12/2025

National University of Singapore (NUS)

- School of Computing

BS in Data Science and Analytics (Honours, Distinction)

08/2020 – 05/2024

National University of Singapore (NUS)

- Faculty of Science
- 2nd Major in Economics, Minor in Computer Science
- NUS Science & Technology Undergraduate Scholarship (MOE Full scholarship)
- Thesis: Semiparametric copula-based model for heavy-tailed financial density estimation